

Lesson 16

Topic: Advanced Charting Techniques

Prerequisites: Download Card_data.csv file and disable Auto DateTime options by Options in Current File and create new DATE table by using Calendarauto and Format functions.

1. **Total Card Limit by Card Brand and Card Type (Stacked Column Chart)**
 - **X-axis:** card_brand (e.g., Visa, Mastercard, Amex)
 - **Y-axis:** Sum of card_limit
 - **Legend:** card_type (e.g., Credit, Debit, Prepaid)
 - **Tooltip:** client_id count (i.e., number of clients using this brand/type combo)

Prerequisites Setup

Before creating the chart:

1. **Load Card_data.csv:**
 - Open Power BI Desktop.
 - Go to **Home > Get Data > Text/CSV** and load the Card_data.csv.
2. **Disable Auto DateTime:**
 - File > Options and settings > Options.
 - In **Current File > Data Load**, uncheck **Auto Date/Time for new files**.
3. **Create a custom Date Table (if needed):**

If your dataset includes a date field (like issue_date), create a proper Date table:

DAX

DateTable =

ADDCOLUMNS(

CALENDARAUTO(),

"Year", YEAR([Date]),

"Month", FORMAT([Date], "MMM"),

"MonthNo", MONTH([Date])
)

- Mark it as a **Date Table**: Go to the Model view → Right-click on DateTable → "Mark as Date Table" → Choose Date.

Stacked Column Chart: Total Card Limit by Card Brand and Card Type

1. Create the Chart

- Go to the **Report** page.
- Choose the **Stacked Column Chart** visual.

2. Set the Fields:

Section Field

X-axis card_brand

Y-axis card_limit → set to **Sum**

Legend card_type

Tooltip client_id → change aggregation to **Count**

Optional: Rename the measure in tooltip to something meaningful like **Client Count**.

3. Customize the Visual (Optional but Recommended):

- **Sort** by card_brand or card_limit.
- **Data labels**: Enable from the **Format Pane**.
- **Colors**: Customize based on card_type for better visual distinction.
- **Title**: Change chart title to something like “Total Card Limit by Brand and Type”.

Extra Tips (Advanced Touches)

- Create a measure for **client count**:

DAX

Client Count = DISTINCTCOUNT(Card_data[client_id])

Add this to the **Tooltip** section for a reusable and optimized calculation.

- Use slicers (e.g., by year or region) to enhance interactivity.

2. Drill Down into Monthly Trends (Drill Down Feature)

- Create a drill-down chart for analyzing card issues by time:
 - Hierarchy: Year → Month (from account_opened_date)
 - Values: Count of card_number
 - Chart type: Stacked Column Chart
 - Enable drill down/up to move between years and months.

Step-by-Step: Drill Down Chart for Monthly Trends

Step 1: Create a Clean Date Column

Since acct_open_date is in MM/YYYY format, we need to convert it into a proper Date.

In Power Query (Transform Data):

1. Select acct_open_date.
2. Add a custom column:

powerquery

```
acct_open_date_clean = Date.FromText("01/" & [acct_open_date])
```

This assumes every date is the **first day of the month**.

3. Change this column type to **Date**.
4. Click **Close & Apply**.

Step 2: Create a Date Table

In **Data view**, create a Date table:

DAX

CopyEdit

DateTable =

ADDCOLUMNS(

```
    CALENDAR(MIN('Card_data'[acct_open_date_clean]),  
    MAX('Card_data'[acct_open_date_clean])),
```

```
"Year", YEAR([Date]),  
"Month Name", FORMAT([Date], "MMM"),  
"Month No", MONTH([Date])  
)
```

Step 3: Create a Relationship

Go to **Model View**:

- Connect DataTable[Date] → Card_data[acct_open_date_clean].

Step 4: Build the Drill Down Chart

1. Insert a **Stacked Column Chart**.
2. **X-axis**:
 - Drag in DataTable[Year], then DataTable[Month Name].
 - Power BI will create a hierarchy automatically: Year > Month Name.

If it doesn't create it automatically:

- Click "..."> **New hierarchy**, add Year and Month Name manually.
3. **Y-axis (Values)**:
 - Drag in card_number (set to **Count**).
 4. **Legend (Optional)**:
 - Add card_type or card_brand for color segmentation.

Step 5: Enable Drill Down / Up

- In the chart toolbar (top-right corner of the visual), you'll see:
 - **Drill Up** (↑),
 - **Go to Next Level** (↗),
 - **Expand All Down One Level** (↘),
 - **Drill Mode** (●) → Toggle this ON.

Now, clicking on a year will **drill into the months** of that year.

Outcome

You now have an interactive **drill-down chart** showing:

- **Years and Months** based on acct_open_date
- Count of **cards issued (card_number)** each month
- Optional: Color-coded by card_type or card_brand

3. **Top 10 Clients by Total Card Limit (Bar Chart)**

- **Axis:** client_id
- **Value:** Total card_limit
- **Filter:** Top 10 clients
- **Sort:** Descending by total card limit
- **Use a bar chart to clearly show top clients.**

Step 1: Clean and Prepare card_limit Column

If not already cleaned:

1. Go to **Power Query (Transform Data)**.
2. Select credit_limit column → **Replace Values:**
 - Replace \$ with nothing (blank).
3. Change data type to **Decimal Number**.
4. Click **Close & Apply**.

You now have a clean numeric column: credit_limit.

Step 2: Create the Bar Chart

1. In **Report View**, insert a **Bar Chart** (horizontal column chart).
2. Drag in fields:
 - **X-axis** → client_id
 - **Y-axis (Values)** → Sum of credit_limit

Step 3: Filter Top 10 Clients

1. Click the **visual**.
2. In the **Filters pane**, under client_id, do the following:

- Click the dropdown → **Filter type: Top N**
- **Show items: Top 10**
- **By value:** Select **Sum of credit_limit**
- Click **Apply filter**

Step 4: Sort the Chart

1. Click on the visual.
2. Click the **ellipsis (⋮)** or dropdown on the chart's top-right corner.
3. Select **Sort by** → **Sum of credit_limit**
4. Order: **Descending**

Step 5: Customize (Optional but Recommended)

- Turn on **Data labels** (Format pane).
- Change the title to: **"Top 10 Clients by Card Limit"**
- Color bars by brand or card type if needed (add to legend).

Final Outcome:

- You now have a **clean bar chart** showing the **top 10 clients by total card limit**, sorted in descending order.

4. Client Drill-through Details

- **Create a drill-through page showing all card details (e.g., card_type, card_brand, limit) for a selected client_id when a user clicks from any chart.**
- **Add filters like card_type or expire_dates (Year).**

Step 1: Create a New Report Page

1. Click the **"+" (New Page)** at the bottom of Power BI.
2. Rename the new page to something like **"Client Details"**.

Step 2: Enable Drill-through by client_id

1. Go to the **Visualizations Pane** → **Drill-through** section (at the bottom).
2. Drag client_id into the **Drill-through** field well.
3. This tells Power BI: “Filter this page based on the selected client_id”.

Step 3: Add a Card Table Showing Card Details

1. Insert a **Table visual** on the page.
2. Add these fields to the table:
 - card_brand
 - card_type
 - card_number (optional — hide partially if needed)
 - credit_limit
 - expires
 - has_chip
 - acct_open_date

Step 4: Add Optional Filters

To allow filtering on this drill-through page:

a. Card Type Slicer

- Insert a slicer visual.
- Add field: card_type

b. Expire Year Slicer

You need to extract year from the expires field.

If expires is a text like "12/2022":

1. Create a **calculated column**:

DAX

Expire_Year = RIGHT('Card_data'[expires], 4)

2. Add this Expire_Year to a slicer.

Step 5: Test the Drill-through

1. Go back to your main page (e.g., Top 10 Clients bar chart).
2. **Right-click** on any bar (i.e., on a client).
3. Choose **Drill-through → Client Details**.
4. You'll be taken to the "Client Details" page filtered to that specific client.

Optional: Add Back Button

1. On the drill-through page:
 - Insert a **Back Button**: Insert → Button → **Back**.
 - It lets users go back to the previous page easily.

Result

- A dedicated page with full card details for each client.
- Interactive filters for card type and expiration year.
- Works from **any visual containing client_id**.

5. Heatmap-style Matrix of Expiry Trends

- **Rows:** card_brand
- **Columns:** Year of expire_dates
- **Values:** Count of cards expiring
- **Format as a matrix with conditional formatting to highlight peaks (red for more expiries).**

Goal: Heatmap Matrix – Expiring Cards by Brand & Year

Step 1: Extract Expiry Year

If your expires column is text like "12/2022", you need to extract the **year**.

Option A: Create a Calculated Column

In **Data view**, create this column:

DAX

```
Expire_Year = RIGHT('Card_data'[expires], 4)
```

This extracts "2022" from "12/2022".

🔧 If expires is already a Date column, use:

DAX

```
Expire_Year = YEAR('Card_data'[expires])
```

Step 2: Create the Matrix Visual

1. Insert a **Matrix** visual.
2. Drag in fields:
 - **Rows** → card_brand
 - **Columns** → Expire_Year
 - **Values** → card_number (set to **Count**)

You can also use client_id for distinct count if needed.

Step 3: Add Heatmap-style Conditional Formatting

1. Click the **Values** section in the matrix.
2. Click the dropdown arrow (▼) next to the value field → **Conditional formatting** → **Background color**.
3. In the dialog:
 - **Based on field:** Count of card_number (or your chosen value)
 - **Minimum color:** Light (e.g., white or light yellow)
 - **Maximum color:** Red (for peak expiry counts)
 - Click **OK**

Now the cells will turn **red** for high expiry counts, creating a heatmap effect.

Optional Formatting

- Turn **column headers** and **row headers** to bold.
- Adjust **gridlines** and **font size** for readability.
- Add a **Title**: "Card Expiry Trends by Brand & Year".

Final Result:

You now have an **interactive heatmap matrix** showing:

- Card **brands as rows**,

- **Expiry years as columns,**
- **Count of expiring cards** with red highlights for peaks.

6. Create a dynamic slicer which will give options to select or to enter number and bar chart or column chart need to show only Top N (N should be selected by user in dynamic slicer) Card Brands according to their card limit. (column chart)

- **X axis: Card Brand**
- **Y axis: Sum of Card Limit**

Step 1: Create a What-If Parameter for N

1. Go to **Modeling > New Parameter > Fields:**

- Name: TopN
- Data Type: Whole Number
- Minimum: 1
- Maximum: 20 (or more)
- Increment: 1
- Check: "Add slicer to this page"

This creates a new table named **TopN** with a column called TopN[TopN Value].

Step 2: Create a Measure to Get Top N Card Brands

Now create the following **measure**:

DAX

TopN Card Limit Filter =

VAR N = SELECTEDVALUE('TopN'[TopN Value], 5)

VAR RankedBrands =

ADDCOLUMNS(

SUMMARIZE('Card_data', 'Card_data'[card_brand]),

```

        "TotalLimit", SUM('Card_data'[credit_limit])
    )
VAR TopBrands =
    TOPN(N, RankedBrands, [TotalLimit], DESC)
RETURN
    IF(
        COUNTROWS(
            FILTER(
                TopBrands,
                'Card_data'[card_brand] = EARLIER('Card_data'[card_brand])
            )
        ) > 0,
        1,
        0
    )

```

Step 3: Create the Column Chart

1. Insert a **Column Chart**.
2. Set fields:
 - **X-axis:** card_brand
 - **Y-axis:** Sum of credit_limit
3. Add TopN Card Limit Filter to the **Visual Filters** pane.
 - Set filter to **"is 1"** to keep only Top N brands.

Step 4: Format & Test

- Place the TopN slicer beside the chart.
- Change its orientation (horizontal/vertical).
- Change the title of the slicer to: **"Select Top N Card Brands"**.
- Sort chart by **Sum of credit_limit**, descending.

Final Result

Now your users can:

- Use a slicer to **dynamically choose how many top brands (N)** they want to see.
- View a column chart that **updates in real time** with the Top N card brands by total credit limit.

7. Download sales.csv file and disable Auto DateTime options by Options in Current File and create new DATE table by using Calendarauto and Format functions.

- Calculate Average number of days between Sales date and previous sales date for each customer
(ex:
last sales date 05.05.2025,
prev sales date 05.01.2025,
sales before prev sales date 04.21.2025
Answer: $7 (4 + 10) / 2$)

Objective:

For each customer, calculate the **average number of days between sales**.

→ Example:

Sales dates: 04/21, 05/01, 05/05 → gaps: 10 and 4 → $\text{avg} = (10+4)/2 = 7$

Step-by-Step Instructions

Step 1: Load sales.csv

1. Go to **Home > Get Data > Text/CSV** → load the sales.csv file.
2. Click **Transform Data** and ensure:
 - SalesDate is of **Date** type.
 - CustomerID (or client_id) is available and clean.

Step 2: Disable Auto DateTime

- File > **Options and settings > Options**
- Under **Current File > Data Load**

- Uncheck "**Auto Date/Time for new files**"

Step 3: Create a Proper Date Table

Go to **Modeling > New Table**, and enter:

DAX

DateTable =

```
ADDCOLUMNS(  
    CALENDARAUTO(),  
    "Year", YEAR([Date]),  
    "Month", FORMAT([Date], "MMM"),  
    "MonthNo", MONTH([Date])  
)
```

Then:

- Right-click the DateTable → **Mark as Date Table** → Select Date.

Step 4: Create Measure for Average Days Between Sales

We'll use **DAX with variables and DATEDIFF** to calculate this:

Option A: Calculated Column (row-by-row)

DAX

CopyEdit

Previous Sale Date =

```
CALCULATE(  
    MAX('sales'[SalesDate]),  
    FILTER(  
        'sales',  
        'sales'[CustomerID] = EARLIER('sales'[CustomerID]) &&  
        'sales'[SalesDate] < EARLIER('sales'[SalesDate])  
    )  
)
```

)

Then add this column:

DAX

Days Since Previous Sale =

DATEDIFF([Previous Sale Date], 'sales'[SalesDate], DAY)

Step 5: Average Days Between Sales (Measure)

Now create this **measure**:

DAX

Average Days Between Sales =

AVERAGEX(

 FILTER(

 'sales',

 NOT(ISBLANK('sales'[Days Since Previous Sale]))

),

 'sales'[Days Since Previous Sale]

)

Final Usage

- Use this **measure** in a Table visual with CustomerID.
- You'll see **average number of days between sales** per customer.
- You can also use a slicer for Year or Month from your DateTable.

Bonus: Format It

You can format the measure to remove decimals:

DAX

ROUND([Average Days Between Sales]